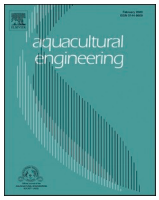
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AI-powered decision support system for mariculture: Real-time fish mortality prediction with random forest

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ABSTRACT

Fish mortality is a significant issue in mariculture, affecting productivity and sustainability. Predicting mortality risk in real-time is crucial for improving decision making and operational efficiency in mariculture management. This paper presents the development of a real-time fish mortality risk prediction model, designed as part of a Decision Support System using the Random Forest machine learning algorithm. The innovative aspect of this study lies in the real-time processing of sensor data to deliver daily mortality risk predictions, allowing for immediate adjustments to management practices. This study integrates water quality parameters (seawater temperature, salinity, conductivity, chlorophyll-a, turbidity, and dissolved oxygen) monitored through a sensor network, with daily fish mortality records input by farmers. The Random Forest model predicted fish mortality risk across five levels with an overall accuracy of 78.6 % and precision exceeding 70 % for each level. The model's feature importance analysis highlights seawater temperature, salinity, and turbidity as key predictors of fish mortality risk. This system supports fish farmers and site managers in daily operational decision making, particularly regarding feed and labor management. Future improvements in data collection and continuous model updates are expected to enhance the accuracy and utility of the Decision Support System in mariculture management.

1. Introduction

Indonesia's vast archipelago, with over 17,000 islands and 81,000 km coastline, hosts one of the world's most dynamic mariculture sectors. Among cultured species, grouper (*Epinephelus sp.*) commands economic importance, accounting for some 11 % of global farmed grouper production and bringing premium prices on export markets (FAO, 2017; Firdaus et al., 2020). Recognizing this economic potential, the Indonesian Ministry of Marine Affairs and Fisheries (MMAF) has strategically supported grouper culture since the late 1990s through the development of research facilities like the Gondol Research Institute for Mariculture (GRIM) in northwestern Bali (MMAF, 2016). This sustained investment has been instrumental in transforming the Gondol area into one of Indonesia's largest and most important grouper-farming centers.

Despite the strategic importance and investment, the high mortality rate of cultured groupers remains a formidable challenge for farmers. Empirical data consistently report mortality rates often exceeding 50 %, with a survey conducted at a prominent grouper mariculture facility in Gondol between 2018 and 2019 revealing an average mortality rate as high as 70 % (Inoue et al., 2020). Yet, the specific underlying environmental drivers have largely remained underexplored. Our research group initiated environmental monitoring in grouper mariculture facilities in 2013 (Wada et al., 2013), partly in response to a hypothesis prevalent among several GRIM researchers and farm managers suggesting that cold water masses causing low dissolved oxygen (DO) level were the primary cause of mass mortality. However, our multi-depth, time series analyses revealed no simple DO and temperature correlation and thus failed to explain the recurrent collapses. This finding underscored the insufficiency of temperature and DO alone to explain the occurrence of grouper mortality and identified a crucial gap in understanding the full spectrum of environmental factors at play. To address this, we subsequently expanded our approach, developing and

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deploying a real-time sensor network capable of monitoring a broader and more comprehensive range of water quality indicators beyond just temperature and DO (Wada et al., 2019).

Beyond the direct economic losses from fish deaths, such high and unpredictable mortality profoundly impacts daily operational logistics, particularly concerning feed and labor management. According to MMAF (2017), grouper feed predominantly consists of wild-caught trash fish, which requires pre-purchase three days in advance and labor-intensive processing (e.g., cutting into smaller pieces) before feeding. The unpredictable nature of mass mortality events severely complicates the precise forecasting of feed requirements and efficient deployment of labor, leading to substantial inefficiencies and economic strain (Firdaus et al., 2020). Consequently, the development of a robust fish mortality prediction scheme, integrated as a decision support system (DSS), is critically needed to enable fish farmers to proactively manage feed procurement and optimize labor allocation, thereby enhancing the economic viability and sustainability of their operations.

Several prior studies have applied binary classifiers through machine learning (ML) and artificial intelligence (AI), such as random forest (RF) and support vector machines (SVM) to predict whether a mortality event will occur under static environmental conditions (Yñiguez and Ottong, 2020; Ewees et al., 2021). Other works have adopted probabilistic anomaly detection and deep learning techniques, including Gaussian models, extreme learning machines, and neural networks to capture more complex, nonlinear interactions among water-quality variables (Lopes et al., 2021; Shi et al., 2024; Ting and Chen, 2024; Wu et al., 2023). However, none offer a real-time, multi-level mortality risk prediction integrated into a daily DSS. For example, (Shi et al., 2024) applied a clustered regularized extreme learning machine to improve short-term DO forecasts in a black bass pond as their first step to predict hypoxia, but their model stops at environmental prediction and does not link forecasts to mortality risk or support real-time decision making. Meanwhile, Yñiguez and Ottong, (2020) utilized a RF model to predict the probability of fish mortality due to harmful algal blooms (HABs) in a Philippine mariculture site, providing a binary mortality (alive or dead) occurrence. Optimized SVM have also been employed for binary mortality prediction of tilapia based on static lab-collected data (Ewees et al., 2021). Lopes et al. (2021) similarly employed a Gaussian probability model for binary prediction of fish deaths exceeding a predefined threshold in indoor tanks. Aravanis et al. (2024) developed ensemble models to predict fish mortality caused by infectious diseases, offering continuous mortality rates, though their application is typically retrospective based on historical data. Deep learning models have also been applied to forecast accidental fish mortality by leveraging water and air quality data with training based on environmental data collected days before a mortality event (Ting and Chen, 2024), thus providing a retrospective analysis of contributing factors rather than real-time forecasting. Studies investigating explainable AI for mortality prediction under specific stressors like ammonia nitrogen similarly utilize historical data to analyze stress-induced mortality (Wu et al., 2023). Previous studies indicated whether mortality is likely to occur or if a threshold is exceeded, rather than offering real-time, quantitative predictions of daily mortality numbers or rates. Furthermore, the integration of these predictive models into a comprehensive and dynamic DSS that provides actionable insights for proactive management, particularly in complex mariculture environments like grouper farming, remains an underexplored area.

Therefore, the objective of this research was to develop an automatically updated ML-based prediction model capable of classifying fish mortality risks into multiple levels over a three-day period. Unlike previous studies that offered binary or retrospective assessments of fish mortality, this study leveraged real-time sensor data to provide actionable predictions, integrating ML and DSS into daily mariculture operations. The model utilizes real-time data collected from a sensor network to monitor water quality (Wada et al., 2019), in conjunction with daily records of fish mortality provided by fish farmers. The goal is to create a robust DSS that can effectively aid fish farmers in their day-to-day operations such as feed and labor management. Furthermore, we also enabled the system to automatically share the computation results with the mariculture site manager and owner through the Internet. This study systematizes information to support the better management of mariculture at the local level.

2. Materials and methods

2.1. Characteristics of floating net cages for grouper mariculture in Gondol

The study was conducted in Penerusan Bay, Gondol, Northwestern Bali, Indonesia (Fig. 1), which served as the study area. Floating net cages are a typical form of mariculture used in fish cultivation. The data collection for this study focused on a specific floating net cage located at the marked location icon in Fig. 1. A detailed aerial image of the floating net cage is shown in Fig. 2(a). The target area consisted of 200 holes within a floating net cage structure. However, fish farmers typically utilize only 70 to 90 % of the available holes, leaving 10 to 30 % empty for rotational cleaning purposes. Rotational cleaning of the net is necessary at least once a month to clean biofouling (MMAF, 2017). The remaining 10 to 30 % of holes are also used for fish quarantine or treatment in cases of disease outbreaks. Depending on the size of the fish, each floating net cage could accommodate approximately 100 to 200 grouper heads. The grouper in the hole of a floating net cage is shown in Fig. 2(b). This scheme represents the standard mariculture practice in Gondol. Once the fry reaches a size exceeding 10 cm, fish farmers often transfer it from the nursery to the floating net cage. Groupers are then regularly sorted and relocated to other holes to reduce size variation within the floating net cage, with the aim of ensuring equal size distribution and minimizing cannibalism and competition for food acquisition (MMAF, 2017). Sorting activities are typically conducted once a week to once every three weeks by fish farmers.

2.2. Real-time water quality data and fish mortality record

This study used a real-time sensor network developed previously by our research group (Wada et al., 2019) to monitor more complex water quality (other than temperature and DO) in the floating net cage. Fig. 3 (a) shows the waterproof housing unit of the sensor network with a control board and a battery inside the housing. Outside, the housing was equipped with solar panels. Maintenance of the housing unit is almost unnecessary because the power comes from the solar panel and its waterproof function. Fig. 3(b) shows the sensors used to monitor water quality data. Seawater temperature, current, salinity, conductivity, chlorophyll, turbidity, and DO. This study monitored the water quality once every 30 min and transmitted the data to the database (DB) server via iridium satellite communication once an hour. To maintain the sensors, fish farmers need to clean the sensors once a week to prevent biofouling. Fig. 3(c) shows a screenshot of the application's data input module for daily grouper mortality.

Fish farmers diligently recorded daily fish mortality and entered the data into a dedicated DB using a specialized iOS application developed for this research. We provided iPhones to the site manager to record the fish mortality. In addition to mortality records, the application allowed for the input of information regarding the timing of sorting activities and the average size of fish, both before and after the sorting process within a specific fish cage hole. However, to maintain focus on and relevance to the research objectives, our analysis concentrated solely on the daily mortality records obtained through this application. It is pertinent to note that the transmission of data from the iOS application to the DB relies upon the utilization of the 3 G/4 G network infrastructure. In instances where signal reception was unavailable, the smartphone device temporarily stored the recorded information until a signal was re-established, facilitating the subsequent transmission of the stored data

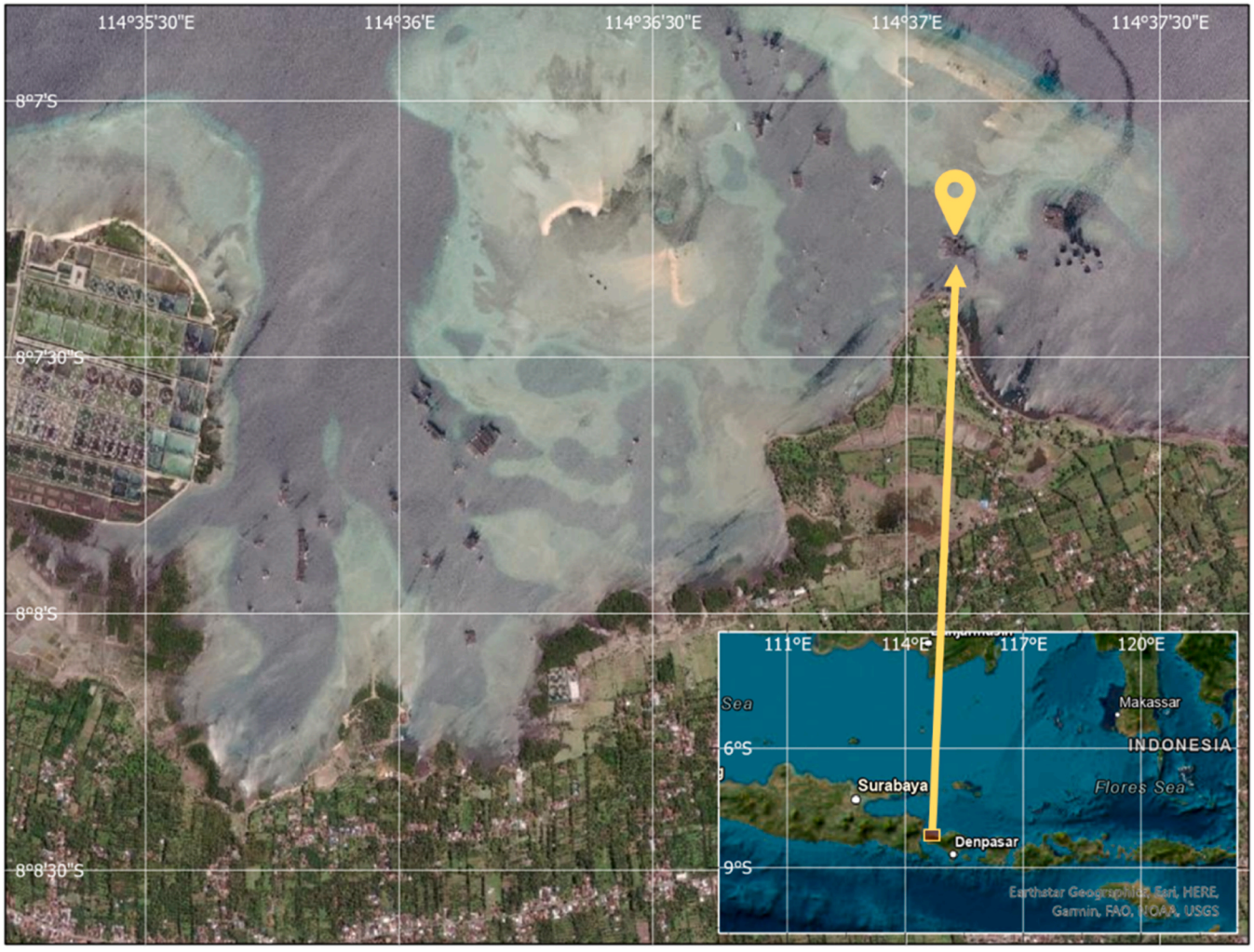


Fig. 1. Location of the target study area.

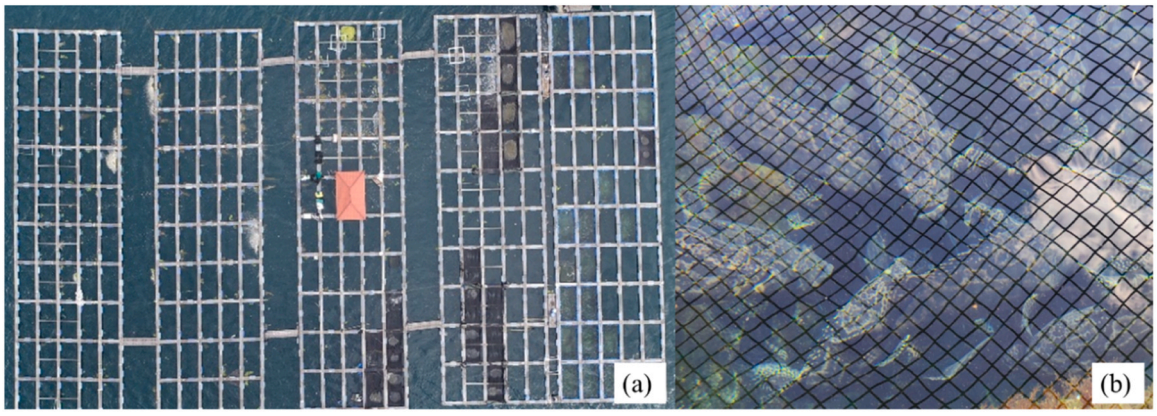


Fig. 2. (a) Aerial picture of a floating net cage set. (b) Grouper within floating net cage.

to the central DB.

The water quality features monitored in this study were based on a literature review. Seawater temperature is generally considered to be one of the most important water quality features for aquaculture (Summerfelt, 2000; Weber et al., 2013; Zhang et al., 2020). Seawater temperature plays an important role in biology by determining the rate of biochemical reactions that eventually affect the food intake of aquaculture fish (Buentello et al., 2000; Goddard, 1996), and it might also affect other water quality features such as chlorophyll-a, pH, and DO concentrations (Weber et al., 2013). The water current is also considered an important factor in the design and operation of aquaculture systems (Liu and Zhang, 2022; Stevens et al., 2008). Johansson et al. (2007) also mentioned that Seawater current helps maintain the water quality and provides the necessary oxygen, nutrients, and waste products, which are critical to the growth and survival of cultured fish. Salinity generally represents the total concentration of dissolved salts in



Fig. 3. (a) Sensor network housing unit. (b) Water quality sensors. (c) Screenshot of application for inputting daily grouper mortality record.

water. Salinity is one of the most important physical features of seawater because it is an essential indicator of climatological change, seawater masses, and ocean dynamics that can affect marine organisms. Salinity is also a key factor in controlling the growth of fish, including cultured fish (Gu et al., 2022; Ytrestøyl et al., 2020). Conductivity serves as a gauge of the collective ionic content within water, indicating its freshness or quality, as suggested by Ogbeibu and Victor (1995). Conductivity is an important consideration for aquaculture sites because it can affect fish growth (Boyd and Lichtkoppler, 1979). Chlorophyll-a is one of the key indicators for assessing water quality because of its response to nutrients and light availability in marine waters (European Environment Agency (EEA), 2022). Chlorophyll-a can indicate a rise in algal biomass, which is considered a sign of deteriorating water quality (Perkins and Underwood, 2000; United States Environmental Protection Agency (USEPA), 2023a). Meanwhile, turbidity is a general term that describes the "cloudiness" or "muddiness" of water, which is measured by the amount of light scattered by the material present in a water sample when a light beam passes through it (United States Geological Survey (USGS), 2018). Turbidity can be used to indicate the presence of algae, pathogens, bacteria, and dissolved organic substances that stain water, colloidal solids, and other contaminants such as lead and mercury (Hamilton and Luffman, 2009; Jung et al., 2014), which affect the growth and survival of fish (Ardjosoediro and Ramnarine, 2002; Boyd and Tucker, 1998). Lastly, DO is a commonly used term describing the quantity of oxygen gas dissolved in water and is essential for the respiration of aquatic animals United States Environmental Protection Agency (USEPA), 2023b). DO is crucial for mariculture because it directly impacts the health and survival of aquatic organisms, including cultured fish, and DO is directly related to the growth and mortality rates of fish (Kim et al., 1995; Mallya, 2007).

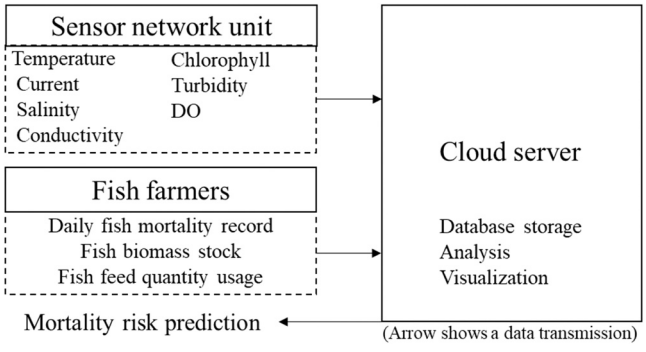


Fig. 4. Diagram of the real-time data collection and transferring scheme in this study.

The data transfer scheme is illustrated in Fig. 4. The cloud server was set up as DB storage, a platform to analyze and construct a fish mortality prediction ML model, and a platform to show the result to fish farmers as well as owners. The automatically monitored data by the sensor network and input by the fish farmers were transmitted to the same DB in the cloud server. The data were then utilized to construct a fish mortality risk prediction model for the next three days. The risk prediction model was updated daily. Next, the results are sent to fish farmers as a DSS for their daily work.

2.3. Fish mortality risk prediction algorithm

Table 1 provides an overview of the data sources, variables, aggregation methods, time spans, and key preprocessing steps used in this study. We combined high-frequency water-quality measurements from our real-time sensor network with daily mortality counts recorded via an iOS application. Sensor network inputs, namely seawater temperature, salinity, conductivity, chlorophyll-a, turbidity, and DO were sampled every 30 min and smoothed using a 3-day moving average (MA) after two-sigma outlier removal. Mortality records were aggregated to daily totals per cage. These two streams were merged into a single machine-learning dataset, labeled into five risk levels via dynamic quartile thresholds. Finally, a time-blocked 90/10 train/test split stratified by risk level ensured balanced evaluation.

The dependent variable consisted of five fish mortality risk levels,

Table 1
Dataset summary.

Data Source	Variables Monitored	Sampling / Aggregation	Time Period	Notes
Sensor network	Seawater temperature, current, salinity, conductivity, chlorophyll-a, turbidity, DO	30 min raw; 3-day moving average	Ongoing since 2021	Two-sigma filter to remove outliers prior to MA
Daily mortality records	Daily count of dead groupers per cage	Daily total	Ongoing since 2021	Entered via iOS app; buffered when offline
Combined ML dataset	3-day MA of each water-quality feature + daily mortality	5-level risk label by quartiles	Ongoing since 2021	Labels: Q1–Q3 + IQR thresholds
Train/Test split	–	90 % train / 10 % test (time-block)	Ongoing since 2021	Stratified by risk level to preserve class balance

calculated from the 3-day moving average of fish mortality, allowing for clear and actionable predictions to be provided to fish farmers and site managers. We designed the five-level risk scale to function analogously to established hazard indices which provide a standardized ordinal framework that enables rapid interpretation of mortality risk and supports timely operational decision making. The five levels of fish mortality were automatically and dynamically computed using quartile and interquartile schemes. Specifically, level 1 was between 0 head mortality of groupers to less than the first quartile. Level 2 was between the first quartile and lower than the second quartile. Level 3 was between the second quartile and less than the third quartile. Level 4 was between the third quartile and less than the 1.5 interquartile. Level 5 was when the 3-day MA of fish mortality exceeded 1.5 interquartile. With this level division method, the level might change every day, depending on the number of fish mortalities.

Raw sensor measurements are subject to occasional errors or anomalies due to biofouling, environmental disturbances, or network issues (Teh et al., 2020). Therefore, preprocessing is required to detect and exclude such data. In this study, we used the two-sigma rule as an omission scheme, which is a statistical method. The two-sigma rule was employed for data filtering to remove outliers, which is a standard approach in time-series analysis to prevent noisy data from affecting the model. Alternative methods, such as median filtering were considered, but not implemented because of the need to retain data variability. All data monitored by the sensor network were used to compute sigma (standard deviation). The computed standard deviation was then used to apply the two-sigma rule for data filtering.

The exact effect of water quality on fish mortality remains unclear. Therefore, we applied a moving average method to better capture these trends in the data. The base number for the moving average is also unknown; hence, we conducted a Fast Fourier Transform (FFT) to detect periodic fluctuations (Edelman et al., 1998) in seawater temperature, one of the generally known variables that affect fish conditions. The FFT was used to identify periodic patterns in seawater temperature fluctuations, enabling the selection of an appropriate window size for the moving average. This approach was chosen due to its effectiveness in isolating frequency components in environmental data. Using FFT, we found that a 3-day period is an important periodic indicator through. Thus, we used a 3-day MA based on the FFT results to capture the trend in the data. The independent variables for the prediction were the 3-day MA of seawater temperature, current, salinity, chlorophyll, turbidity, and DO.

We did not perform a formal multicollinearity test prior to training the RF model because ensemble tree methods are generally robust to correlated inputs and will still identify optimal split points even when predictors share information (Breiman, 2001). However, high collinearity can obscure the interpretation of feature importance, strongly correlated variables tend to share importance, making attribution less clear (Strobl et al., 2007). To mitigate this effect, we excluded conductivity as an independent variable because of its high correlation with salinity. Moreover, we also did not conduct sensitivity analysis because the RF's built-in permutation importance measures provide an effective proxy for quantifying each feature's influence on model predictions (Scikit-Learn, 2022).

This study linked the daily fish mortality record with screened water quality data to develop a three-day fish mortality prediction model as a DSS for fish farmers. The DSS used in this study uses ML, specifically an RF algorithm. RF is a widely used ML technique for prediction and classification (Biau and Scornet, 2016; Speiser et al., 2019). It is a supervised ML technique based on an ensemble method that uses multiple learning algorithms to obtain better predictive performance than that obtained from any of the constituent learning algorithms alone (Rokach, 2010). Moreover, RF was selected because of its ability to handle complex interactions between environmental factors and its robustness against overfitting in small datasets compared to alternative algorithms such as SVM or neural networks (Adugna et al., 2022; Boateng et al., 2020). RF is constructed using multiple decision tree learning algorithms, resulting in a forest of trees, and hence the name random forest (Breiman, 2001). This technique also corrects the decision tree's tendency to overfit the training set. Another advantage of the RF model was its feature importance score. The feature importance score in an RF model indicates the extent to which each input variable (or feature) contributes to the predictive power of the model (Menze et al., 2009). This score quantifies the significance of each feature in enhancing the accuracy of the predictions. RF was implemented using Scikit-learn library version 1.1. Hyperparameters such as the number of decision trees, maximum tree depth, and minimum samples per leaf in the algorithm were tuned using a trial-and-error approach, and performance was evaluated using cross-validation to optimize predictive accuracy. This method is often necessary, as experimentation is crucial for discovering the optimal settings for a specific dataset (Brownlee, 2016; Goodfellow et al., 2016). The hyperparameter tuning process was guided by the recommendations provided in the Scikit-learn library documentation (Scikit-Learn, 2022), with careful consideration given to avoid overfitting and underfitting. Although we evaluated multiple hyperparameter configurations (including varying n_estimators, max_depth, and min_samples_split) via grid search and cross-validation, the optimal setup for our dataset proved to be Scikit-Learn's default settings combined with random_state = 42. Specifically, we used RandomForestClassifier(random_state = 42) with n_estimators = 100, max_depth = None, min_samples_split = 2, min_samples_leaf = 1, max_features = 'auto', and bootstrap = True.

The dataset comprised 1300 fish-mortality records collected from June 10, 2021 through December 31, 2024. Although data collection is ongoing, the December 2024 cutoff ensures a sufficiently long time series for robust model training and evaluation. We employed a stratified random split of 90 % for training (n = 1170) and 10 % for testing (n = 130) to ensure robust model assessment while retaining the majority of observations for learning complex patterns. This 90/10 division is a widely used compromise in applied ML, in particular the large training set allows the RF to capture subtle relationships among water-quality predictors and mortality outcomes, while the held-out 10 % provides an unbiased estimate of true generalization performance. Within this development set, we applied 10-fold cross validation to tune and select hyperparameters. This 10-fold scheme provides a low-bias estimate of generalization error while maintaining sufficient data per fold for stable evaluation (Kuhn and Johnson, 2013).

We evaluated fish mortality risk prediction using accuracy and precision (Brownlee, 2018; Tharwat, 2020). Accuracy is defined by Eq. (1),

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \tag{1}$$

However, the accuracy is often considered insufficient for comprehensively evaluating the prediction model. Consequently, the combination of accuracy and precision was ideal. Precision is defined in Eq. (2),

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

where TP denotes true positives and FP denotes false positives. TP is an outcome in which the model correctly predicts the positive category. FP is an outcome in which the model incorrectly predicts the positive category. In other words, precision is a ratio that quantifies the number of positive class predictions that actually belong to the positive group.

Fig. 5 illustrates the workflow of our real-time fish mortality–risk prediction system. First, an in situ sensor network deployed on floating net cages continuously measures water quality. These raw readings are transmitted hourly via satellite to a centralized cloud database. Upon arrival, a two-sigma filter removes irregular values before computing three-day MA for each water quality variable. Simultaneously, daily mortality counts entered by site managers through a mobile application are aggregated into three-day moving averages and combined with the

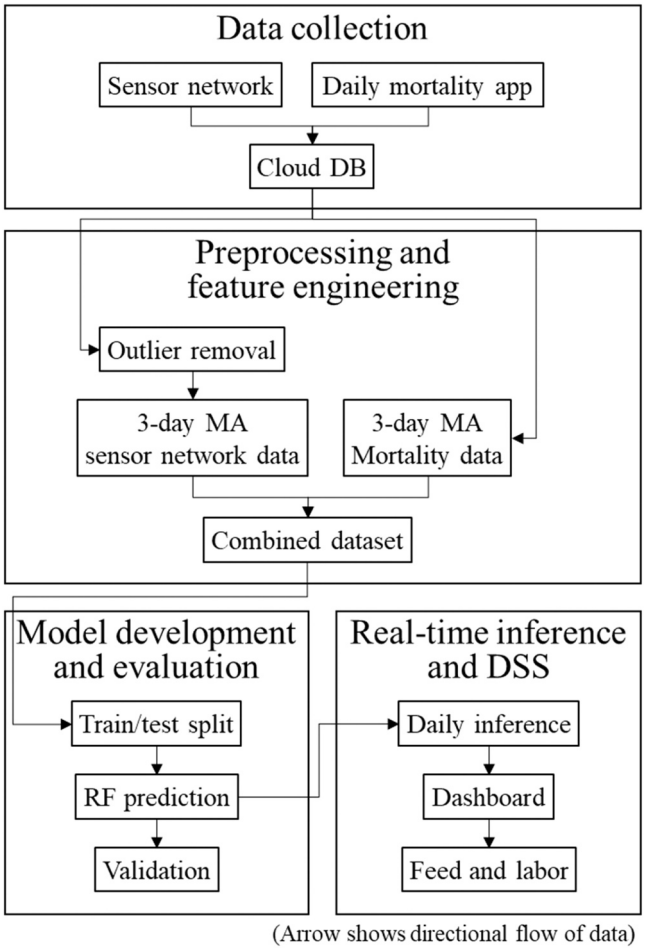


Fig. 5. Workflow of the real-time fish mortality–risk prediction system.

water-quality features to form a structured ML dataset. Each night at 01:00 AM, the dataset is fed into a pre-trained RF classifier, which outputs a five-level mortality risk prediction for the next three days. Finally, the risk levels and underlying feature importances are published to the decision support dashboard, where site managers can review actionable forecasts and adjust feeding and labor plans accordingly.

All model training, evaluation, and data processing were conducted on our dedicated Linux server (Debian 10.9, Python 3.11). This server, which also hosts the live database, is housed at the MMAF office in Jakarta to ensure long-term operational continuity. Its specifications are Intel Xeon 16-core CPUs (2.8 GHz), 2 × NVIDIA Tesla T4 GPUs, 8 × 32 GB DDR4 RAM modules (256 GB total), and 8 × 1.6 TB SSDs. This machine supports both RF computations and real-time data ingestion for our DSS.

3. Results and discussions

3.1. Fish mortality risk prediction model performance

We applied the RF algorithm to predict fish mortality risk into five risk levels for the next three days. After running the algorithm 10 times, we found that the average accuracy was 78.6 %. Table 2 shows a sample confusion matrix comparing the predicted risk level and the actual risk level in this study. Level 1 indicates a low level of fish mortality risk (ranging from 0 fish deaths to only a few fish deaths that can be counted on one hand), while Level 5 represents a high level of fish mortality risk (where dozens or even hundreds of fish are likely to die, with the risk level dynamically changing based on the daily fish mortality record). The model demonstrated high precision for predicting low (Level 1, 89 %) and medium (Level 3, 93 %) fish mortality risks, suggesting that it reliably identifies minimal and moderate risks. However, the precision decreases for Levels 2 (70 %) and 4 (72 %), which reflect moderate-risk scenarios, indicating some room for improvement in these areas. Importantly, the model showed relatively lower precision (71 %) for predicting the highest risk level (Level 5), which may be due to the critical nature of accurately identifying high-risk situations to prevent large-scale fish mortality. The model exhibited lower precision in predicting high mortality risk (Level 5), which may be attributed to the limited instances of high-risk conditions in the dataset or overlapping feature values across risk levels. In addition, there are noticeable misclassifications, particularly between adjacent risk levels, such as Level 5 being misclassified as Level 3. This highlights the model's difficulty in distinguishing between moderate from high-risk scenarios, potentially underestimating severe situations. Improving the accuracy of the model in predicting high-risk levels will ensure timely strategy and decision making in daily mariculture operations.

Table 2
Confusion matrix of fish mortality risk prediction.

Prediction Actual	Level 1	Level 2	Level 3	Level 4	Level 5	Precision
Level 1	**32**	3	1	0	0	89 %
Level 2	2	**26**	8	0	1	70 %
Level 3	0	0	**14**	0	1	93 %
Level 4	0	0	5	**13**	0	72 %
Level 5	2	0	4	1	**17**	71 %

To further analyze the model's discriminative power and the trade-off between true positive rate and false positive rate, as well as precision and recall for each class, we generated Receiver Operating Characteristic (ROC) curves and Precision-Recall (PR) curves, as shown in Fig. 6. The ROC curves in Fig. 6(a) illustrate the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) for each risk level. Level 1 exhibits the highest Area Under the Curve (AUC) of 0.94, indicating robust separability and effective performance in identifying low-risk conditions. Conversely, Level 5 shows the lowest AUC of 0.71, confirming the model's reduced ability to distinguish high-mortality events from other classes. The PR curves in Fig. 6(b) further highlight this matter, with Level 1 achieving an Average Precision (AP) of 0.90, while Level 5 has an AP of 0.73. The relatively lower AP values for Levels 2, 4, and 5 (0.74, 0.75, and 0.73, respectively) align with the lower precision observed in the confusion matrix, suggesting that the model struggles more with these intermediate and high-risk categories. This is often indicative of an imbalanced dataset where high-risk events are less frequent, making it harder for the model to learn their distinct characteristics.

To provide statistical evidence for the overlapping feature values and the distribution of data across risk levels, Fig. 7 presents the 3-day MA distributions for seawater temperature, turbidity, and salinity across the five mortality risk levels. As shown in Fig. 7, the distributions for each risk level, particularly for Levels 2, 3, and 4, exhibit significant overlap across all three key environmental parameters (temperature, turbidity, and salinity). For instance, in seawater temperature (Fig. 7(a)), while Level 1 (low risk) is predominantly associated with lower temperatures and Level 5 (high risk) with higher temperatures, the distributions for moderate levels (2, 3, 4) show considerable overlap, making it challenging for the model to draw clear boundaries. Similarly, for turbidity (Fig. 7(b)) and salinity (Fig. 7(c)), the density plots show that a given range of these parameters can correspond to multiple risk levels. This visual evidence directly supports the hypothesis that overlapping feature values contribute to the misclassifications observed in the confusion matrix, particularly for adjacent risk levels. Furthermore, the density of the curves, especially for Level 5, appears lower and broader compared to Level 1, suggesting fewer data instances for these high-risk events. This relative scarcity of high-mortality data points makes it more

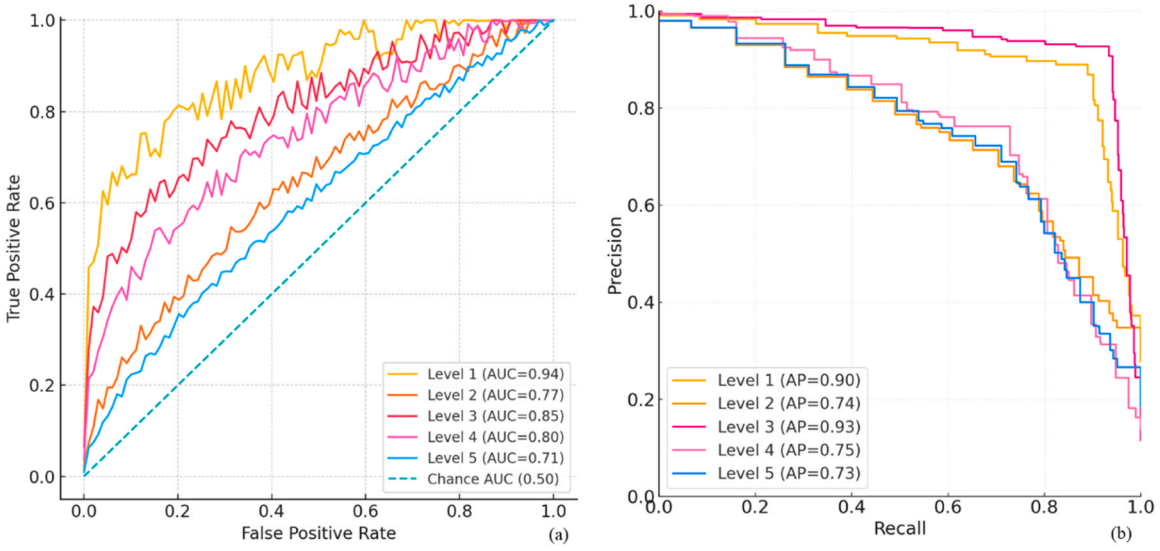


Fig. 6. (a) Multi-class ROC curves. (b) Multi-class PR curves.

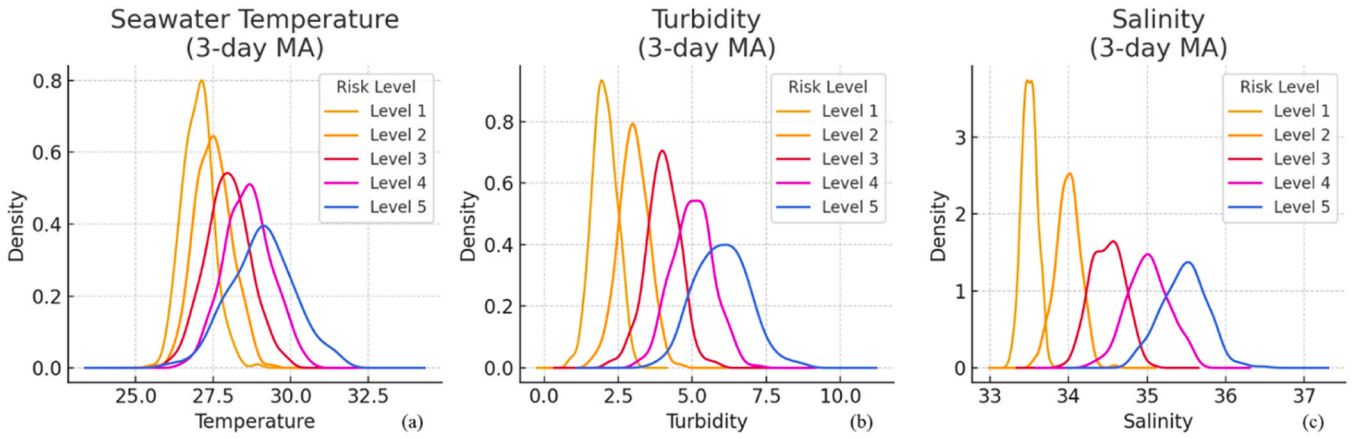


Fig. 7. 3-day moving-average distributions for (a) seawater temperature, (b) turbidity, and (c) salinity across the five risk levels.

difficult for the RF model to learn robust and precise decision boundaries for Level 5, leading to the observed lower precision and AUC for this critical class. This imbalance in the dataset, coupled with the inherent overlap in environmental conditions across different mortality outcomes, underscores the complexity of predicting high-risk events and highlights areas for future data collection and model refinement.

3.2. *Fish mortality predictor importance*

In addition, we investigated the factors influencing fish mortality risk and found that seawater temperature, salinity, and turbidity were key variables for predicting fish mortality in the target area, based on the RF feature importance output shown in Fig. 8. With a feature importance score of 0.2938, temperature emerged as the most influential factor for predicting fish mortality risk. This indicates that fluctuations in water temperature play a crucial role in fish survival, which aligns with the general understanding that temperature significantly affects fish physiology and behavior. These findings are consistent with those of previous studies showing the effects of seawater temperature on fish mortality (Albert and Ransangan, 2013; Pepin, 1991; Saunders et al., 1975). Seawater temperature, turbidity, and salinity ranked next, with feature importance scores of 0.1948 and 0.1899, respectively. Turbidity may affect fish health by impairing their ability to feed or breathe (Bonner and Wilde, 2002), whereas salinity influences osmotic balance, which is

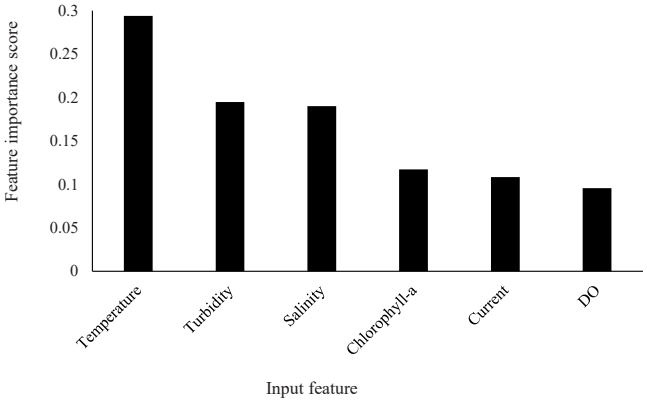


Fig. 8. RF feature importance.

a critical factor for fish survival in marine environments (Morgan and Iwama, 1991). In contrast, chlorophyll-a, current, and DO had lower feature importance scores but remained relevant variables. The importance of seawater temperature aligns with its known impact on fish metabolic rates, while turbidity likely influences visibility and feeding efficiency, contributing to increased mortality risk. The relative

importance of these factors suggests that continuous monitoring and control of these variables is essential for effectively managing fish health and mortality in mariculture. These results align with the ecological expectations, confirming that environmental conditions such as temperature and water quality parameters are directly linked to fish mortality outcomes.

Although chlorophyll-a and DO are recognized as critical drivers of fish health and pond productivity (Perkins and Underwood, (2000); (United States Environmental Protection Agency (United States Environmental Protection Agency (USEPA), 2023b)), they emerged with comparatively low importance in our Random Forest model. One likely explanation is that, at the Gondol mariculture site, both parameters remained largely within narrow, non-stressful ranges throughout the monitoring period. In this open-water system, tidal flushing effectively moderates phytoplankton biomass, with our chlorophyll-a sensors recorded values consistently below 3 μg/L, ensuring chlorophyll-a levels rarely exceed thresholds associated with hypoxic events or harmful algal blooms (Raulo et al., 2025). Similarly, DO sensors recorded values consistently above 6 mg/L, above the 5 mg/L lower limit recommended for grouper culture (MMAF, 2017). If chlorophyll-a and DO levels remained consistently within optimal or non-critical ranges for the majority of the study period, their variability might not have been a strong discriminating factor for mortality events in this specific dataset. In such a scenario, the RF predictioin model would naturally assign lower importance to features that do not frequently fluctuate into mortality-inducing thresholds. Therefore, while chlorophyll-a and DO are undeniably vital for fish health, their lower feature importance in this specific RF model reflects their predictive contribution within the context of our particular dataset and the observed environmental conditions at the study site, rather than negating their overall biological significance.

However, owing to the limited amount of continuous data available, environmental factors may have interacted, leading to compound effects that were not fully captured in this study. As the amount of data increases, it is likely that the differences in the relevance of RF feature importance scores will become more distinct, providing a clearer understanding of how each environmental factor contributes to fish mortality risk. A limitation of this study is the reliance on sensor data, which are prone to errors due to biofouling or environmental disturbances. In the future, it is crucial to encourage fish farmers to regularly perform sensor cleaning and maintenance to ensure more accurate data collection. Accurate and precise predictions based on these improved data are expected to be incorporated into a DSS to aid fish farmers in their daily operations.

3.3. *Real-time DSS dashboard*

Fig. 9 shows a screenshot of the website we developed to provide feedback to fish farmers or mariculture site managers. To access information on the website, users must log in with the account credentials we created for them. After logging in, a greeting along with the user's account name is displayed in the top-right corner of the page. The red box at the top of the screen shows mortality risk prediction, which serves as part of the DSS. We automated the process for mortality risk prediction,

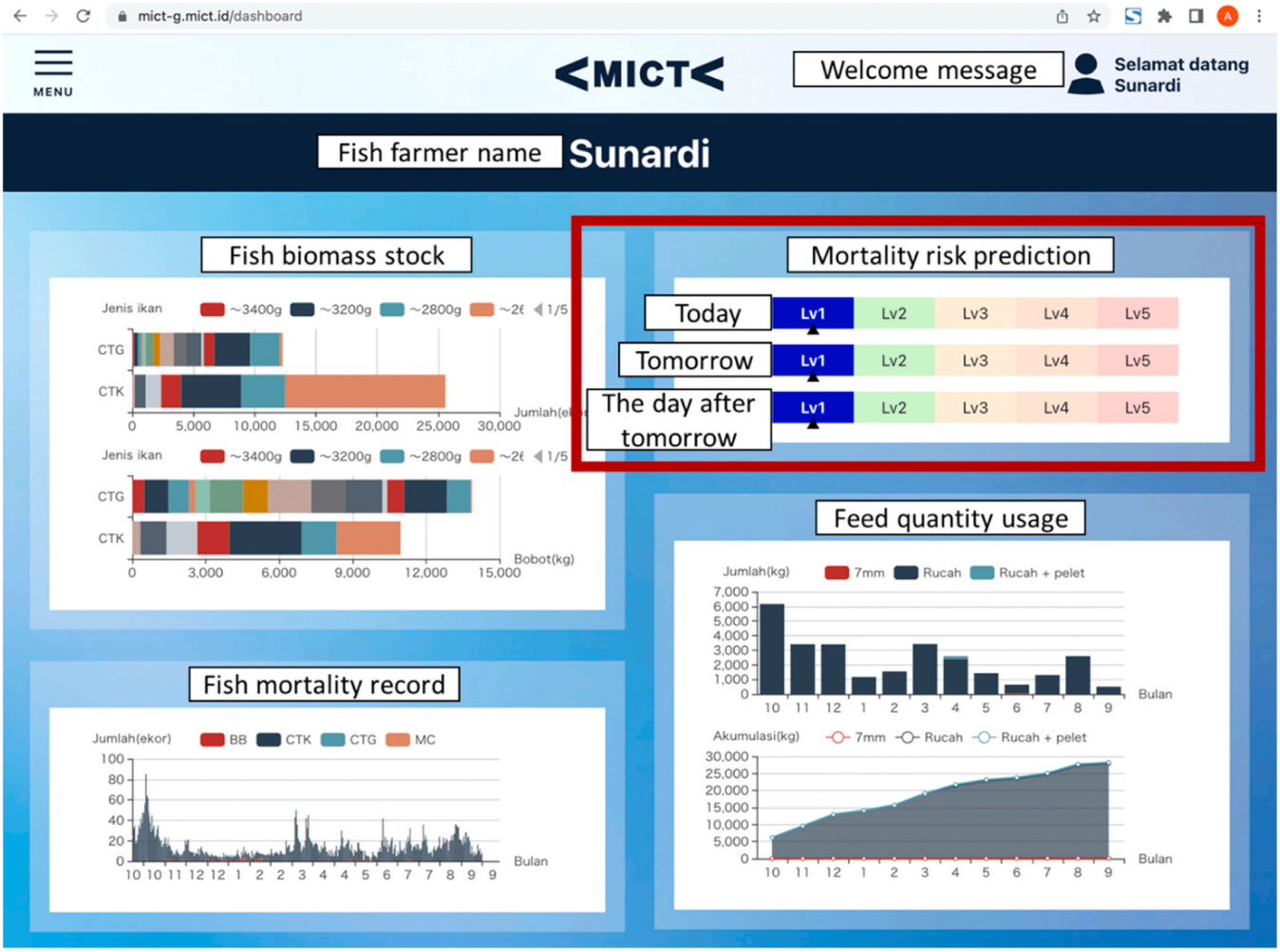


Fig. 9. Realtime DSS dashboard interface for fish farmers and site managers.

including data accumulation, mortality risk level updates, and machine learning model updates, which ran daily at 01:00 AM. The prediction results were provided for three days: today, tomorrow, and the day after tomorrow. These prediction outcomes can serve as a reference for site managers or fish farmers in planning daily operations, such as labor and feed management. As mentioned in the introduction, this prediction can help determine the quantity of fish feed required (e.g., fish trash for feed) and effectively distribute labor. In addition to mortality risk prediction, the website provides records of fish biomass stock, fish mortality, and feed quantity usage, all of which serve as essential references for DSS.

4. Conclusions

This paper presents the development of an AI-powered DSS for mariculture, featuring an ML model for predicting fish mortality risk over a three-day period. By integrating real-time water quality data from a sensor network with daily fish mortality records, our developed RF prediction model achieved an average accuracy of 78.6 % and precision exceeding 70 % across five dynamically defined risk levels. The feature importance analysis highlighted seawater temperature, salinity, and turbidity as key environmental indicators influencing mortality in the target area. This system represents a significant step towards enabling proactive management in aquaculture, providing fish farmers and site managers with actionable insights for optimizing daily operations, such as feed quantity and labor distribution.

Beyond these empirical results, our work carries several practical implications. As the first pilot implementation of comprehensive, sensor-based mortality monitoring for grouper in Indonesia, the platform has been deployed on a dedicated server housed within the MMAF office in Jakarta. This close partnership with the MMAF ensures system maintenance, data stewardship, and long-term operational support as one of the key factors for real-world adoption. Moreover, by delivering daily risk forecasts through a user-friendly web dashboard, we have laid the groundwork for active stakeholder engagement. While systematic feedback collection from site managers and farmers is only now beginning, preliminary responses suggest the potential to fine-tune alert thresholds and feed management protocols based on user experience.

Several limitations and avenues for further development remain. First, questions of scalability and cost-effectiveness, such as the unit cost of sensor maintenance that require formal economic evaluation. Second, the long-term reliability of the model under sensor drift and biofouling has yet to be quantified. In particular, periodic retraining schedules and automated sensor health diagnostics will be critical components of future system upgrades. Third, while this pilot has demonstrated reproducibility within a single site, extending the approach to other species, cage designs, and hydrographic regimes will test the generality of both the RF algorithm and our preprocessing pipelines. To address these challenges without delaying immediate benefits, we propose a staged rollout: (1) continue gathering user feedback to refine the dashboard interface and alert logic; (2) conduct cost-benefit analysis and sensor-network optimization studies; and (3) develop automated model retraining workflows tied to data availability triggers.

In conclusion, this research provides a robust proof-of-concept for an AI-powered DSS capable of real-time fish mortality prediction, offering tangible benefits for mariculture management. The identified areas for future work are not indicative of incomplete study but rather represent the logical and necessary next steps for translating this promising pilot into a widely adopted, resilient, and economically impactful solution for the global aquaculture industry. Continued collaboration with governmental bodies and direct engagement with stakeholders will be key to realizing the full transformative potential of such intelligent systems.

CRediT authorship contribution statement

Ramadhona Saville: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization, Writing – review & editing. **Atsushi Fujiwara:** Visualization, Validation, Software, Data curation. **Katsumori Hatanaka:** Validation, Software, Project administration, Methodology, Funding acquisition, Conceptualization. **Masaaki Wada:** Validation, Software, Project administration, Methodology, Funding acquisition. **Aris Yaman:** Validation, Methodology, Formal analysis. **Reny Puspasari:** Methodology, Investigation. **Hatim Albasri:** Methodology, Investigation. **Nugroho Dwiyoga:** Methodology, Investigation.

Author statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He/she is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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